

ULTRASONIC
WIND SENSOR
LCJ CAPTEURS

CV7-E

Ultrasonic Wind Sensor



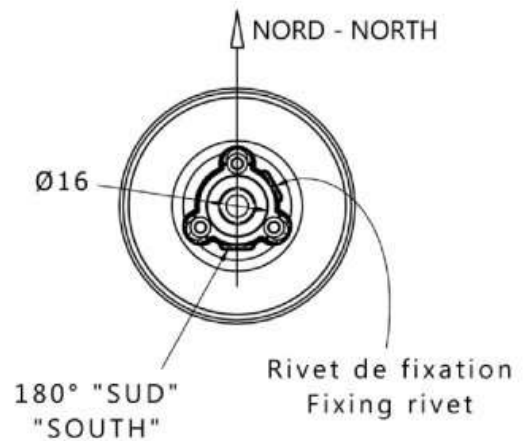
1. Installation

Prepare the mast head for the mechanical installation. Use the template for the bracket's screwing. Align the bracket in order to have the CV7-E sensor heading to the bow while the mast is aligned to the North.

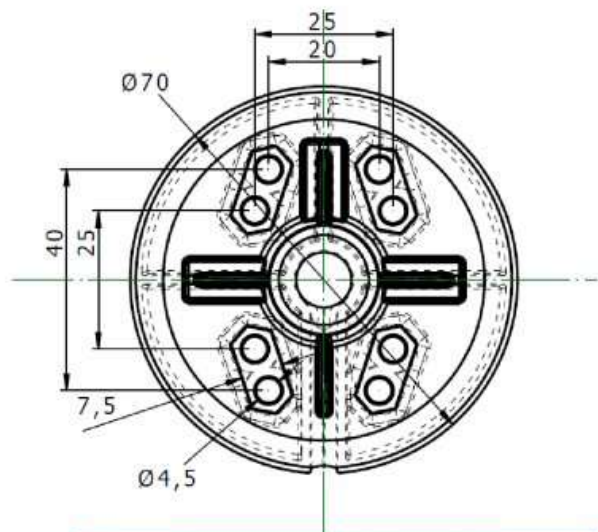
The sensor's fine alignment will be obtained with the software calibration application.

Run the cable down to the display's interface, avoiding locations where other cables or equipment may produce electromagnetic interferences.

Sensor view from below



Bracket mounting template



Note : for mounting on vertical surface, the brackets RA106BRACKET and RA106BRACKINOX from Glomex® are compatible. (see the example picture, not supplied)



2. Wind sensor alignment and configuration

CV7-E is delivered with the following configuration :

NMEA0183®

Filtering: 2 s

Forces of wind unity : Nds

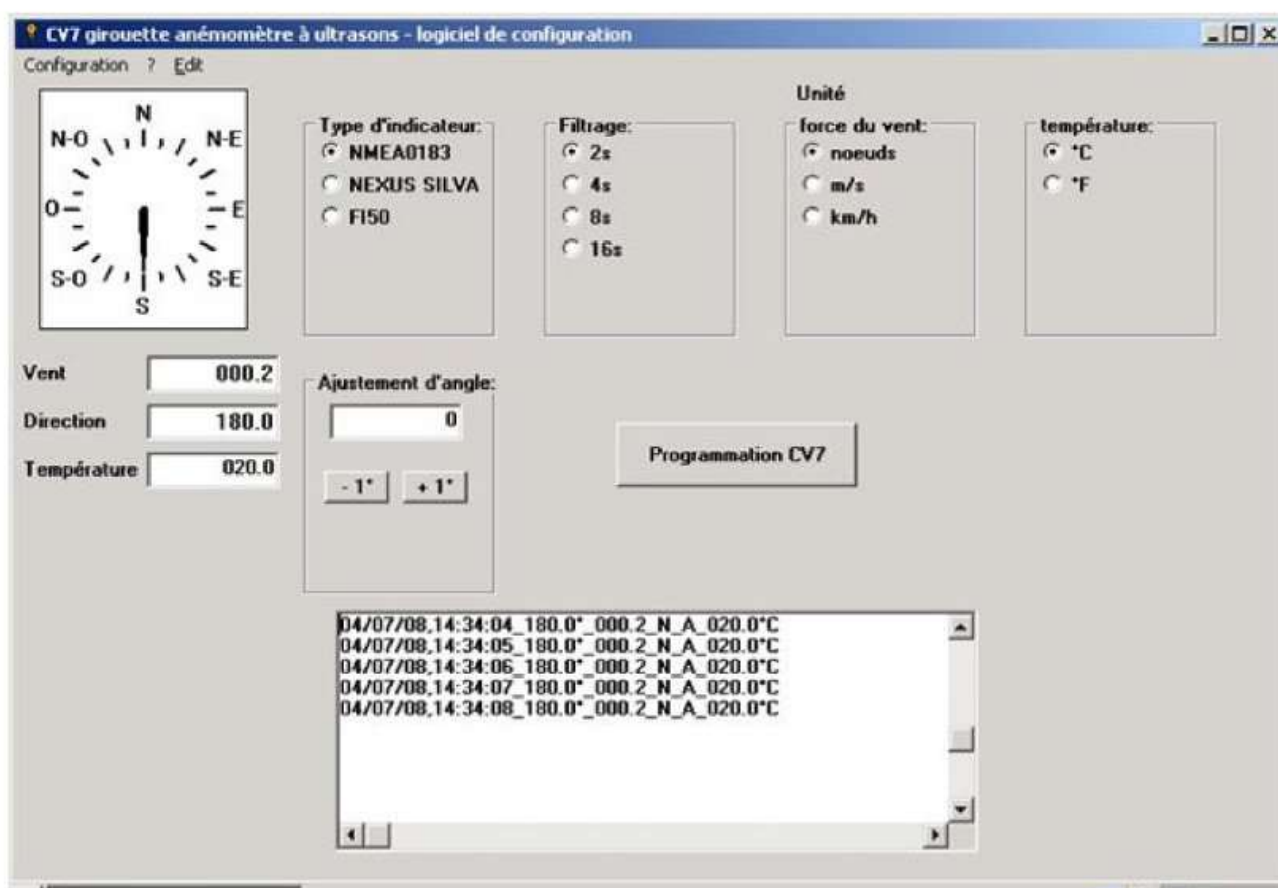
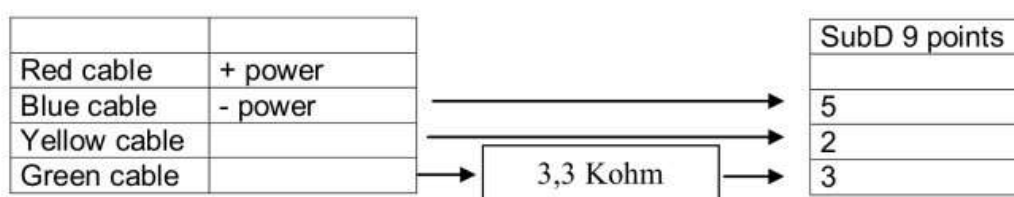
Temperature unity: °C

Adjustment of the angle of the wind: 0°

To choose other parameters, use the downloadable software on this website:

www.lcjcpteurs.com. « Configuration_CV7-C ».

Realise the following connexions (only for configuration):

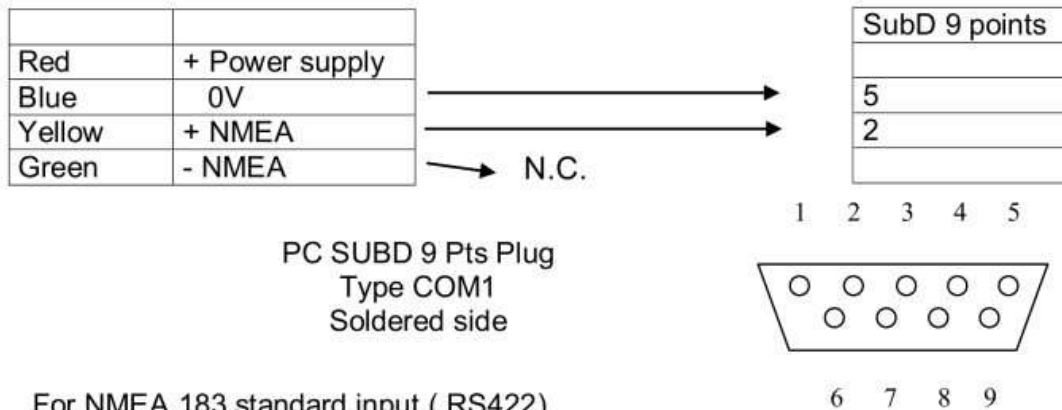


3. Interconnections NMEA 0183

Realise the following connexions:

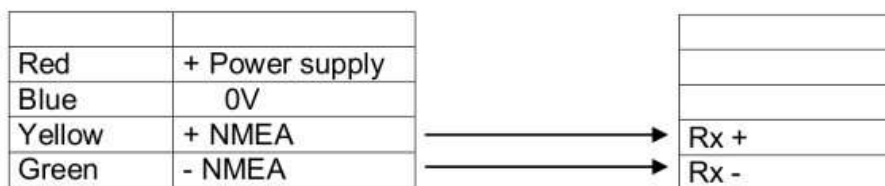
- For RS232 Type PC input

Power supply connections 12V (8 à 30 VDC). Wire Blue - / Wire Red +

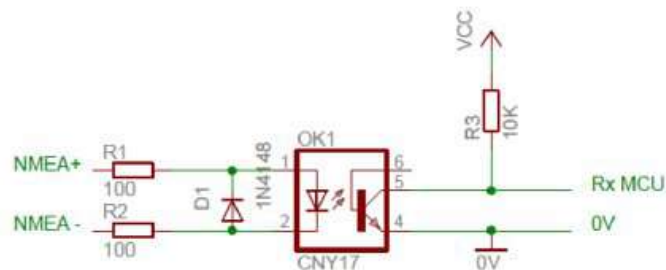


- For NMEA 183 standard input (RS422)

Power supply connections 12V (8 à 30 VDC). Wire Blue - / Wire Red +



Typical diagram NMEA0183 to use along cable.



4. CV7-E Technical Specifications:

Wind speed range	0.25-40 m/s (0.5 – 80 knots)
Wind speed resolution	0.1 m/s (0.1 knots)
Wind speed sensitivity	0.13 m/s (0.25 knots) Independent of weather conditions
Wind direction range	0-359 °
Wind direction resolution	1°
Wind direction sensitivity	+/- 1 °
Power requirement	8 V to 33 V DC
Power consumption	9 mA
Dimensions / Sensor	Cylinder Ø 64 mm - height 61 mm
Boot up delay	4 seconds
Cable	25 m standard delivery 4x0.22 mm ² . Ø4 mm
Arm	Aluminium, 300 mm
Cable weight	20 gr/m
Weight / Sensor + arm and base	200 gr
Weight / Sensor	100 gr
Operational temperature	-15 °C to +55°C (without iceing)
Environement	Sensor IP67
CE Mark	EN 55022 EN 55024



Typical wind tunnel report (see : http://www.lcjcpteurs.com/know/quality_anemometer/?lang=en)

wind tunnel speed: 5.2 Hz = 5.9 knots - test temperature: 20°C
Measuring wind speed depending on its direction

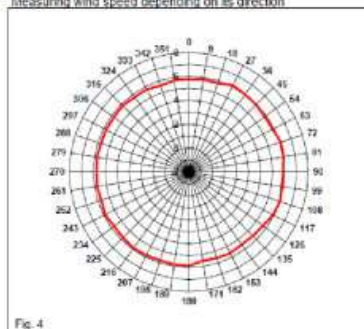


Fig. 4
mean force: 5.60 Nds
sigma force: 0.14 Nds

angle to direction deviation

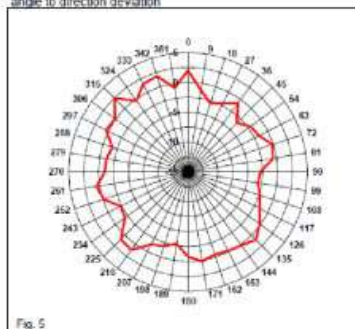


Fig. 5
mean angle deviation: -0.74 °
sigma angle deviation: 1.48 °

wind tunnel speed: 11.4 Hz = 15.0 knots - test temperature: 20°C
Measuring wind speed depending on its direction

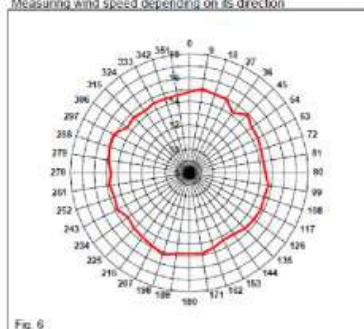


Fig. 6
mean force: 14.67 Nds
sigma force: 0.24 Nds

angle to direction deviation

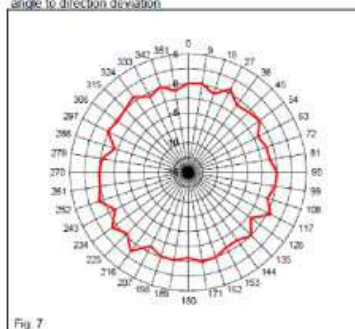


Fig. 7
mean angle deviation: -0.33 °
sigma angle deviation: 0.77 °

5. NMEA0183 protocol:

Baud rate 4800, data bit 8, parity none, stop bit 1

Flux every 266 milli-second

Variable length fields, delimiter « , ».

Start of sentence : \$

End of sentence : CR,LF

Voltage outputs: 0 or 5V

Symmetrical outputs

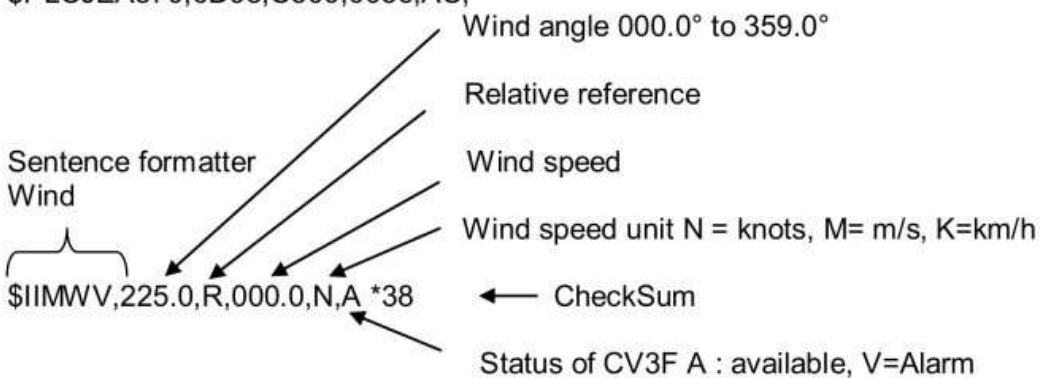
Examples :

\$IIMWV,225.0,R,000.0,N,A*38

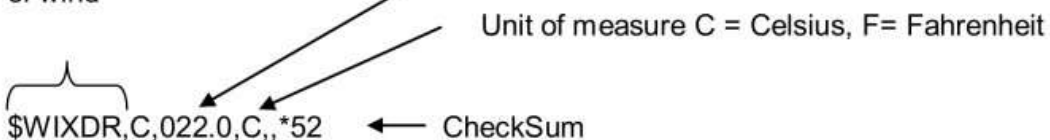
\$WIXDR,C,022.0,C,,*52

\$PLCJ,5801,5F01,AA,4253,3341

\$PLCJEA870,6D98,C500,0056,AC,



Sentence formatter
Temperature
of wind

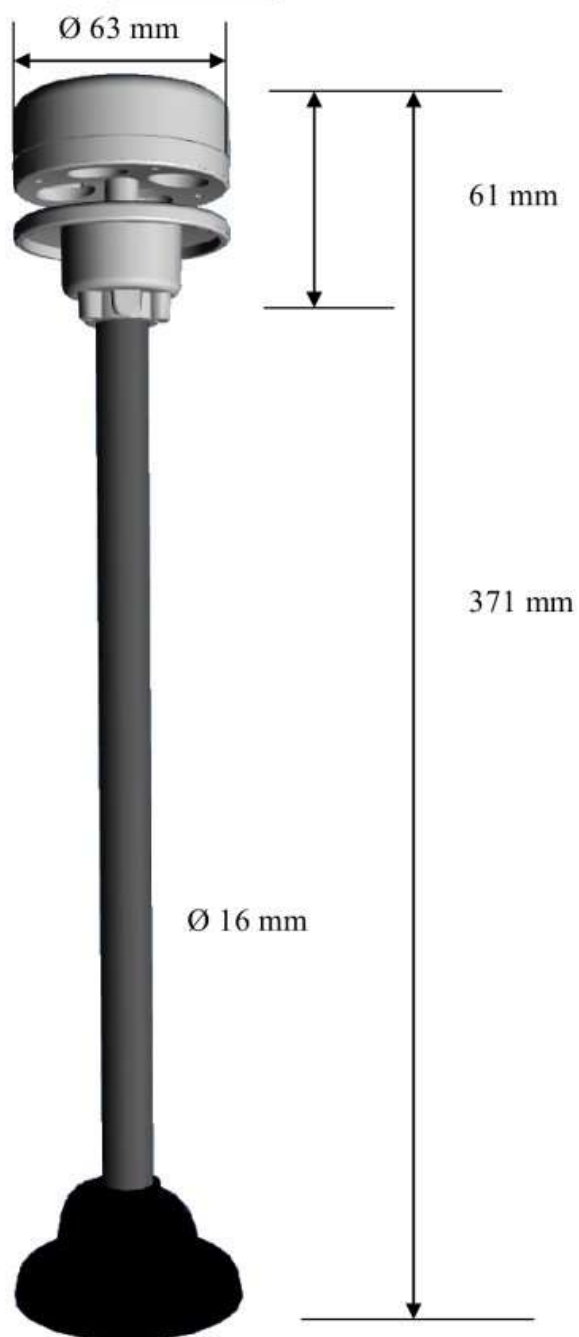


\$PLCJ,AAAA,BBBB,CC,,,DDDD,EEEE

\$PLCJEFFFF,GGGG,HHHH,IIII,JJ

} Sentences for LCJ CAPTEURS
technical service

6. Dimensions



7. **Warranty**

Your CV7-Eis warranted against manufacturing defects in materials and workmanship for a period of 2 years from the date of purchase.

LCJ Capteurs will at its discretion, repair or replace faulty product free of charge excluding the cost of shipping. Proof of purchase shall be required in order for a warranty claim. All claims shall be made in writing to LCJ Capteurs and once it is accepted by LCJ Capteurs, the faulty product must be sent to LCJ Capteurs address in Vertou, France.

For further information, please visit www.lcjcpteurs.com where video presentations are available.



EN 55022 EN 55024



Remember to respect the local regulations: dispose non-working electronic equipment to an appropriate waste disposal center.